4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

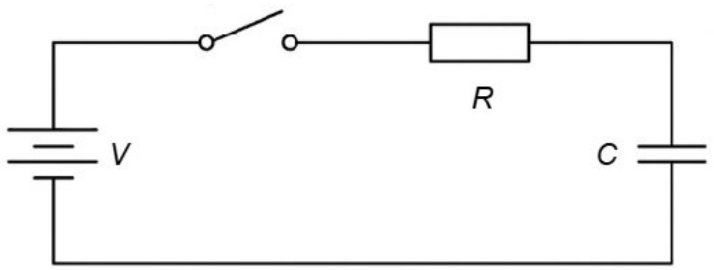


Fig. 4.1

(a) Calculate the time constant for the circuit.

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

current = ... mA [2]

[Total: 5]